

PROJECTS

Here are the projects that I am involved in here at IU, School of Informatics.



Advisor: [Prof. Haixu Tang](#)

Project1. An Energy Based Motif Finding Algorithm Using Chromatin Immunoprecipitation on Microarray Data.

This project aims at developing an improved energy based motif finding algorithm utilizing ChIP-on-Chip data.

- Quantitative Modeling of Protein-DNA Interaction Using ChIP-on-Chip Data
 - [Presentation at Emerging Technology Workshop \(CGB\)](#) (Mar. 23, 2005)
 - [Term Project Presentation \(L529\)](#) (Apr. 25, 2005)
 - [Progress Report](#) as a research article (L519) (Dec. 17, 2004)
 - [Final Report](#) as a research article (L529) (May. 7, 2005)

Project2. Differential Location Analysis of Transcription Factors.

This project aims at revealing the underlying correlation among eukaryotic transcription factors by integrative utilization of ChIP-on-Chip, Microarray, PPI data. This project is thesis project.

- [Presentation](#) at [COL group research meeting](#) (Oct. 3, 2005)

- [Capstone Project Proposal](#) (Draft) (Oct. 15, 2005)



[Andrew-Dalkilic Group](#) (Prof. Justen Andrew and Prof. [Mehmet Dalkilic](#))

Project1. IndiGene : Integration and Discovery in Gene Network

Functional genomic data have created a need for tools to integrate and analyze gene networks within a single framework. We focus on the model *Drosophila melanogaster* from a systems biology perspective, using a logic-theoretic (extensional) framework to combine disparate data sources with the goal of discovering gene function. In this framework, we define genes as externally annotated units (from FlyBase), and their interactions are defined as functional connections in a common genetic pathway, or which participate in a common process. Multiple sources of data such as microarray coexpression, protein-protein interaction, genetic interaction, transcription factor binding site overlap, and RNAi phenotype, are combined. These overlaps both support inferences from individual data sets, and further refines groups of genes into coherent clusters. For this integration, we use simple deductive rules based on expert knowledge of the data used, and allow a wide latitude for the user to set thresholds of belief. The system is also readily extensible to new sources of data. Using these techniques, we find that new functional connections can be made between genes because of the multiple types of information used. Using GO annotation, we find significant enrichment for clusters of functionally related genes, demonstrating that the approach taken has successfully integrated disparate genomic-scale data. An illustration is given imputing a tentative biological process for two proteins with no known function.

- [Presentation File for PPI with all papers included](#) (ZIP)(Apr 15, 2005)
- [Integration of Protein-Protein Interaction Data](#) (ZIP) (May 13,2005)

- [Related PPI and Protein Information Sources](#)
- POSTER: [InGene : Integrated Gene Network](#)
- PAPER: [Indigene : Integration and Discovery in Gene Networks](#) (Submitted to 2005 RECOMB Satellite Workshops on Regulatory Genomics and Systems Biology)